UE26CY151A · ENGINEERING CHEMISTRY

SEMESTER 1 · UNIT 1

05

QUICK REVISION

# Short Notes & Keywords

Exam-ready short notes, a keyword index and a last-ten-minutes checklist.

M1
**Molecular Spectroscopy**
8 topics

M2
**Phase Equilibria**
5 topics

M3
**Computational Chemistry**
1 topic

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|---|---|
| **Exam weight** | ★★★ high · ★★☆ medium · ★☆☆ low |
| **Flags** | NOT ON CURRENT SLIDES · TRANSCRIPT ONLY · ADDED FOR CLARITY |
| **Sources** | S004·8 slide 8 of deck 004 · T009 class transcript · PDF Unit-1 slide PDF · ESA past paper |

Compiled from PESU AV-summary transcripts, class slides and past ESA papers · 13 October 2026

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## M1 Molecular Spectroscopy

T1 EM radiation & interaction with matter ★★★

T2 Electromagnetic spectrum & regions ★★★

T3 Born–Oppenheimer approximation ★★☆

T3b Beer–Lambert law ★☆☆

T4 Rotational (microwave) spectroscopy ★★★

T5 Vibrational (IR) – SHO ★★★

T6 Vibrational – anharmonic ★★☆

T7 Electronic spectroscopy & Franck–Condon ★★★

## M2 Phase Equilibria

T8 Basic terms & phase rule ★★★

T9 Derivation of phase rule ★★★

T10 One-component system: water ★★★

T11 Two-component: reduced phase rule & Pb–Ag ★★★

T12 Thermal analysis / cooling curves ★★☆

## M3 Computational Chemistry

T13 Computational chemistry ★☆☆

\+ Keyword index

\+ Last 10 minutes before the exam

MODULE 1

# Molecular Spectroscopy

T1 · EM radiation & interaction with matter | T2 · Electromagnetic spectrum & regions | T3 · Born–Oppenheimer approximation | T3b · Beer–Lambert law | T4 · Rotational (microwave) spectroscopy | T5 · Vibrational (IR) – SHO | T6 · Vibrational – anharmonic | T7 · Electronic spectroscopy & Franck–Condon

## T1 EM radiation & interaction with matter ★★★

**SHORT NOTE**

Spectroscopy studies radiation–matter interaction. A spectrophotometer has source, monochromator, sample, detector, recorder. EM waves have E ⊥ B ⊥ direction. $E = h\nu$, $\nu = c/\lambda$, $\bar{\nu} = 1/\lambda$. Atomic spectra are lines; molecular spectra are bands.

- **absorption spectrum** — radiation absorbed as molecule moves to higher level
- **band spectrum** — molecular spectrum from electronic+vib+rot transitions
- **wavenumber** — $1/\lambda$ in $cm^{-1}$

## T2 Electromagnetic spectrum & regions ★★★

**SHORT NOTE**

Radio → NMR/ESR; microwave → rotation; IR → vibration; UV-Vis → valence electrons; X-ray → inner electrons; γ → nuclei. Raman: Rayleigh (no shift), Stokes (lower ν), anti-Stokes (higher ν).

- **Rayleigh scattering** — elastic scattering
- **Stokes line** — Raman line at lower frequency
- **anti-Stokes line** — Raman line at higher frequency

## T3 Born–Oppenheimer approximation ★★☆

**SHORT NOTE**

$E = E_{rot} + E_{vib} + E_{elec}$, $E_{rot} \ll E_{vib} \ll E_{elec}$, components independent; translation negligible.

## T3b Beer–Lambert law ★☆☆

**SHORT NOTE**

A = log($I_0$/I) = εcl (older syllabus; not on current slides).

## T4 Rotational (microwave) spectroscopy ★★★

**SHORT NOTE**

Rigid rotor: two masses at fixed $r_0$. I = $\mu r_0^2$. $\varepsilon_J$ = BJ(J+1). Needs permanent dipole; ΔJ=±1. Lines at 2B(J+1), spacing 2B. From B get I and $r_0$.

- **centre of mass** — point where $m_1 r_1 = m_2 r_2$
- **gross selection rule** — dipole requirement
- **moment of inertia** — $I = \mu r_0^2$
- **reduced mass** — $\mu = m_1 m_2/(m_1+m_2)$
- **rigid rotor** — fixed-$r_0$ model
- **rotational constant** — $B = h/8\pi^2 Ic$

## T5 Vibrational (IR) – SHO ★★★

**SHORT NOTE**

SHO: spring/Hooke's law, parabolic PE. $\varepsilon_v = (v+½)\bar{\nu}$, ZPE = ½$\bar{\nu}$, equal spacing, Δv=±1 → one line at $\bar{\nu}$. Needs changing dipole; $\bar{\nu} = (1/2\pi c)\sqrt{(k/\mu)}$.

- **Hooke's law** — F = −kx
- **force constant** — k, N $m^{-1}$
- **zero-point energy** — ½$\bar{\nu}_{osc}$

## T6 Vibrational – anharmonic ★★☆

**SHORT NOTE**

Morse potential: $\varepsilon_v = (v+½)\bar{\omega_e} - (v+½)^2\bar{\omega_e}x_e$. Levels converge; Δv=±1,±2...; fundamental, overtones; hot bands left of fundamental.

- **Morse function** — anharmonic PE
- **anharmonicity constant $x_e$** — small positive constant in Morse levels
- **fundamental** — v=0→1
- **hot band** — transition from v=1

- **overtone** — v=0→2, 3…

## T7 Electronic spectroscopy & Franck–Condon ★★★

**SHORT NOTE**

HOMO→LUMO transitions; vibrational coarse structure gives progressions (v′,v″); no vibrational selection rule; Franck–Condon: vertical transitions → intensity pattern by $r_e'$ vs $r_e''$; continuum = dissociation.

- **HOMO / LUMO** — frontier orbitals
- **coarse structure** — vibrational structure on electronic bands
- **continuum** — non-quantised dissociation absorption
- **fine structure** — rotational structure
- **progression** — series (v′,0) lines
- **vertical transition** — Franck–Condon

MODULE 2

# Phase Equilibria

T8 · Basic terms & phase rule | T9 · Derivation of phase rule | T10 · One-component system: water | T11 · Two-component: reduced phase rule & Pb–Ag | T12 · Thermal analysis / cooling curves

## T8 Basic terms & phase rule ★★★

SHORT NOTE

Phase = homogeneous, separable part; component = min. independent constituents; F = min. intensive variables. F = C−P+2.

- **Gibbs phase rule** — F = C−P+2
- **bivariant** — F = 2
- **component** — min. independent constituents
- **degrees of freedom** — F
- **invariant** — F = 0
- **phase** — homogeneous separable part
- **univariant** — F = 1

## T9 Derivation of phase rule ★★★

SHORT NOTE

Variables P(C−1)+2, constraints C(P−1), equal μ in all phases → F = C−P+2.

- **chemical potential** — $(\partial G/\partial n_i)$; equal in all phases at equilibrium

## T10 One-component system: water ★★★

SHORT NOTE

F = 3−P. Areas ice/water/vapour F=2; lines OA, OB, OC F=1; triple point O F=0 (0.0098 °C, 4.58 mmHg); critical 374 °C, 218 atm; OC slope negative; OA′ metastable.

- **Clausius–Clapeyron** — dP/dT = ΔH/TΔV
- **critical point** — 374 °C, 218 atm
- **metastable** — supercooled water, OA′
- **triple point** — 0.0098 °C, 4.58 mmHg

## T11 Two-component: reduced phase rule & Pb–Ag

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SHORT NOTE

Condensed system at fixed P: F = C−P+1. Pb–Ag eutectic: 303 °C, 2.6 % Ag, F=0; curves F=1.

- **condensed system** — no gas phase
- **eutectic** — lowest-melting mixture, F=0
- **reduced phase rule** — F = C−P+1

## T12 Thermal analysis / cooling curves

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SHORT NOTE

Cooling curves: halt for pure/eutectic, slope break for mixtures; plot to build phase diagram.

- **thermal analysis** — cooling-curve study

MODULE 3

# Computational Chemistry

T13 · Computational chemistry

## T13 Computational chemistry

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SHORT NOTE

Computational chemistry simulates molecules: MM/MD (classical), semi-empirical, ab initio, DFT (quantum). Cheap, safe; AI adds speed.

- **DFT** — uses electron density
- **ab initio** — wavefunction-based
- **molecular dynamics** — atomic motion simulation
- **molecular mechanics** — balls and springs
- **semi-empirical** — approximate Schrödinger

# Keyword index

**ab initio** T13
wavefunction-based

**absorption spectrum** T1
radiation absorbed as molecule moves to higher level

**anharmonicity constant $x_e$** T6
small positive constant in Morse levels

**anti-Stokes line** T2
Raman line at higher frequency

**band spectrum** T1
molecular spectrum from electronic+vib+rot transitions

**bivariant** T8
F = 2

**centre of mass** T4
point where $m_1 r_1 = m_2 r_2$

**chemical potential** T9
$(\partial G/\partial n_i)$; equal in all phases at equilibrium

**Clausius–Clapeyron** T10
$dP/dT = \Delta H/T\Delta V$

**coarse structure** T7
vibrational structure on electronic bands

**component** T8
min. independent constituents

**condensed system** T11
no gas phase

**continuum** T7
non-quantised dissociation absorption

**critical point** T10
374 °C, 218 atm

**degrees of freedom** T8
F

**DFT** T13
uses electron density

**eutectic** T11
lowest-melting mixture, F=0

**fine structure** T7
rotational structure

**force constant** T5
k, N $m^{-1}$

**fundamental** T6
v=0 → 1

**Gibbs phase rule** T8
F = C−P+2

**gross selection rule** T4
dipole requirement

**HOMO / LUMO** T7
frontier orbitals

**Hooke's law** T5
$F = -kx$

**hot band** T6
transition from v=1

**invariant** T8
F = 0

**metastable** T10
supercooled water, OA′

**molecular dynamics** T13
atomic motion simulation

**molecular mechanics** T13
balls and springs

**moment of inertia** T4
$I = \mu r_0^2$

**Morse function** T6
anharmonic PE

**overtone** T6
v=0 → 2, 3…

**phase** T8
homogeneous separable part

**progression** T7
series (v′,0) lines

**Rayleigh scattering** T2
elastic scattering

**reduced mass** T4
$\mu = m_1 m_2/(m_1+m_2)$

**reduced phase rule** T11
$F = C-P+1$

**rigid rotor** T4
fixed-$r_0$ model

**rotational constant** T4
$B = h/8\pi^2 Ic$

**semi-empirical** T13
approximate Schrödinger

**Stokes line** T2
Raman line at lower frequency

**thermal analysis** T12
cooling-curve study

**triple point** T10
0.0098 °C, 4.58 mmHg

**univariant** T8
$F = 1$

**vertical transition** T7
Franck–Condon

**wavenumber** T1
$1/\lambda$ in $cm^{-1}$

**zero-point energy** T5
$\frac{1}{2}\bar{\nu}_{osc}$

# Last 10 minutes before the exam

1. $E = h\nu$, $\nu = c/\lambda$, $\bar{\nu} = 1/\lambda$
2. $E_{rot} \ll E_{vib} \ll E_{elec}$ (microwave < IR < UV-Vis)
3. $\varepsilon_J = BJ(J+1)$; lines at $2B(J+1)$; spacing $2B$; first line at $2B$
4. $I = \mu r_o^2 \rightarrow r_o = \sqrt{(I/\mu)}$, $\mu$ in kg
5. Microwave needs a permanent dipole, $\Delta J = \pm 1$
6. $\bar{\nu} = (1/2\pi c)\sqrt{(k/\mu)}$; $\varepsilon_v = (v+½)\bar{\nu}$; ZPE = ½$\bar{\nu}$
7. IR needs a changing dipole; SHO gives one line ($\Delta v = \pm 1$)
8. Fundamental $\bar{\omega}_e(1-2x_e)$; 1st overtone $2\bar{\omega}_e(1-3x_e)$
9. Franck–Condon: vertical transitions; 3 cases; continuum = dissociation
10. $F = C - P + 2$ from $P(C-1)+2 - C(P-1)$
11. Water: $F = 3 - P$; triple point 0.0098 °C, 4.58 mmHg; critical 374 °C, 218 atm
12. OC slope negative because ice → water shrinks ($\Delta V < 0$)
13. Two components at fixed P: $F = C - P + 1$
14. Pb–Ag eutectic 303 °C, 2.6 % Ag, F = 0; Pb 327 °C, Ag 961 °C